

Plan

Begin writing about the existing methods and new methods.

Finish in 1 week

Deadline: June 25th.

2 subsections.

Methods:

Data analysis, introductory.

Results:

Most critical things should be graphs.

Results of new methods, graphs.

Introduction

1. Introduction

1.1 Motivation

Much of human life revolves around communication, and the human voice is among its richest carriers, conveying not only our words, but emotions, thoughts, and intention. Within the roughly 500 milliseconds it takes to say a single “hello,” a listener has already made a stable evaluation of how trustworthy the speaker seems (McAlear, Todorov & Belin, 2014). The judgement is triggered automatically without deliberation, and is assessed to the prosodic properties of the voice, rather than the content of what is said. These rapid impressions of which are involuntarily can directly cause real consequential decision-making: Who passes the interview and gets hired, who gets the loan, or who is believed. Any system that takes voice as an input could thereby be subject to the same signal that human listeners interpret socially. For instance, Voice assistants, customer-service routing, fraud-detection telephony, recruitment screening, and increasingly courtroom and clinical decision-support tools all derive features from speech in which a growing number of decision-support tools attach consequences to them (Castelfranchi et al., 1993). Therefore, a machine-learning model that purports to detect trustworthiness from voice can inherit the social weights of that judgment, along with the risk of being systematically biased.

1.2 What's Known.

Perceived vocal trustworthiness varies systematically with pitch, intonation, speech rate and voice quality (Maltezou-Papastylianou, Scherer & Paulmann, 2025a). Fundamental frequency and its variability shift trustworthiness judgements (McAlear, Todorov & Belin, 2014; Schirmer, Chiu, Lo, Feng & Penney, 2020), intonation contour modulates them (Belin, Boehme & McAlear, 2017), and speech rate together with the harmonics-to-noise ratio, an index of voice quality, contributes further (Schirmer et al., 2020).

A perceptual literature has begun to identify which acoustic properties carry trustworthiness. Perceived vocal trustworthiness tracks a small set of acoustic properties (pitch, intonation, speech rate and voice quality) that listeners appear to weigh consistently (Maltezou-Papastylianou, Scherer & Paulmann, 2025). Higher and more variable fundamental frequency (F0) shifts trust judgements (McAlear, Todorov & Belin, 2014; Schirmer, Chiu, Lo, Feng & Penney, 2020); the shape of the intonation contour carries additional weight (Ponsot et al., 2018); and slower speech rate together with a cleaner harmonics-to-noise ratio (HNR) further modulates how trustworthy a speaker is heard to be (Schirmer et al., 2020). The acoustic correlates of trust, in short, are known.

1.3 The Gap

However, most studies of vocal trustworthiness, and the validation analysis released with the source corpus, pool all speakers and listeners together. Little is known about whether the same acoustic features carry trust across ethnicity, age groups and sexes, and models trained on one demographic may not transfer to another. The work in voice trustworthiness has been largely built on young, white, Western speakers, a demographic skew named as White Western individualistic

bias (WWIB) by Taylor and Rommelfanger (2022). Two problems follow, one of demography and one of method: (i) fairness, in that the existing validation of the Trustworthy Intent in Speech (TIS) Corpus shows tree-based classification accuracy heavily biased toward White demographic groups (71% against 68% and 66%), and (ii) stability, whether a fairness result measured once can survive retraining.

As for stability, fairness metrics are overwhelmingly reported from a single training run, yet deep machine-learning models are sensitive to random initialisations. Therefore, a fairness gap measured once can shift substantially when the same model is retrained under a different seed. Underspecification, the condition in which many models perform near-identically well on the training set, but exhibit unexpected results in real-world domains, is now recognised as a pervasive obstacle to trusting any single evaluation (D'Amour et al., 2020). If the fairness gap of a retrained model can silently triple, a one-off audit is not a safety guarantee. Frame stability is thereby a required property for a deployer or regulatory.

1.4 Our Approach

We address both problems of the Trustworthy intent in Speech Corpus (Maltezou-Papastylianou, Scherer & Paulmann, 2025b), a publicly released dataset of 1,152 utterances from 96 untrained speakers spanning three ethnic backgrounds (White, Black and South Asian), two age groups and both sexes. We evaluate five classifiers on a single 80/20 split of the data, stratified jointly by trustworthy-intent label and ethnicity so that every model is trained and tested on the same partition. The Random Forest and Logistic Regression baselines of the source paper, a three-layer artificial neural network (ANN), a one-dimensional convolutional network (1D-CNN), and DANN-Trust, a domain-adversarial network whose gradient-reversal layer drives the shared encoder to remain predictive of trust while becoming uninformative about ethnicity, following Ganin et al. (2016). Each model is retrained under five random seeds on the fixed split, so that both overall accuracy and the per-ethnicity fairness gap Δ , the difference between the best and worst served ethnic group, are reported as distributions rather than as single points. To ask whether ethnicity information is removed from a representation rather than merely left unused, we probe each frozen encoder with a classifier trained to recover ethnicity from it.

1.5 Findings preview

Our findings form a three-part argument. First, the proposed deep machine-learning models have a higher overall accuracy than the published baselines while more than halving the per-ethnicity gap and raising South Asian accuracy by roughly eight points, with no fairness-specific machinery and purely from non-linear feature interactions. Second, mean fairness alone does not separate the strongest conventional network from DANN-Trust as both settle at essentially the same average gap. So, a single-run comparison would find no reason to prefer one over the other. Third, the multi-seed protocol does separate them as DANN-Trust's fairness gap and accuracy vary far less from seed to seed, and so delivers the predictable retrain behaviour that a single run cannot certify. However, we are careful not to overclaim as the frozen-encoder probe shows that adversarial training reduces the ethnicity information carried in the representation, but does not erase it.

1.6 Contributions

The main contributions of this work are summarized as follows:

A fixed-split, multi-seed evaluation. We fix a single stratified split and retrain every model across five seeds so that fairness is measured as a distribution, and stability becomes a reportable quantity.

Reproduction of the published baselines & A controlled comparison of five classifiers. We re-implement the Random Forest and Logistic Regression baselines of the source paper and evaluate them alongside a three-layer artificial neural network (Dense $128 \rightarrow 64 \rightarrow 32$ with batch normalisation and dropout), a 1D convolutional network (Conv1D-64 $\rightarrow$ Conv1D-32 $\rightarrow$ Dense-64) and DANN-Trust under one identical protocol, enabling like-for-like comparison of accuracy and fairness across model classes.

DANN-Trust. We introduce the capstone of this study, a domain-adversarial network for vocal trust classification whose gradient-reversal layer targets the stability of the fairness gap under retraining rather than peak accuracy on a single run.

A frozen-encoder demographic probe. We provide a probe that quantifies how much ethnicity information remains recoverable from each model's learned representation, distinguishing information that is removed from information that is merely unused.

An open multi-seed fairness analysis. We release per-ethnicity, per-age and per-sex accuracy and AUC, per-seed metrics for every model, and a unified comparison harness that reproduces every number in this paper. The source paper reports only aggregated, single-seed values, so this release supports per-demographic auditing of future models on the same corpus.

1.7 Roadmap

The remainder of the paper is organised as follows. Section II surveys prior work on voice acoustics of trustworthiness, demographic bias in voice AI, and adversarial debiasing. Section III describes the TIS Corpus and our feature pipeline. Section IV details the five classifiers we compare and the evaluation metrics. Section V presents the experimental results. Section VI discusses the fairness–accuracy trade-off and the limitations of our study. Section VII concludes.

Related Work

2. Related Works

2. Acoustic correlates of vocal trustworthiness / social voice perception

Dataset

4. Dataset and Features.

4.1 Dataset description

4.1.1 The Trustworthy Intent in Speech (TIS) Corpus

All experiments conducted are attributed to the Trustworthy Intent in Speech (TIS) Corpus released by Maltezou-Papastylanou, Scherer, and Paulmann [1], an open-access dataset distributed under the CC-BY 4.0 license on the Open Science Framework at doi.org/10.17605/OSF.IO/45D8J [17]. To our knowledge, it is the first publicly available speech-trust dataset that systematically spans multiple ethnic backgrounds, two age groups and both sexes.

The corpus comprises 1,152 utterances produced by 96 untrained adult speakers. Younger speakers and older White speakers were recruited online via the Prolific platform, whereas most older Black and older South Asian speakers were recruited through posters and word of mouth in the United Kingdom, owing to limited responses on Prolific [1]. Each speaker produced the same set of sentences twice: once in a natural voice (labelled *Neutral*) and once while attempting to convey trustworthiness (labelled *Trustworthy*), yielding a binary intent label (Neutral vs. Trustworthy) with 576 utterances per class.

The demographic metadata consist of three factors: ethnicity (White, Black, South Asian), age group (younger adults aged 18 to 45, older adults aged 60 and over), and sex (female, male). Table 1 reproduces the speaker breakdown from the source paper. The four younger-adult ethnic cells are balanced at $N = 20$ each with an approximately even sex split, but the older cells are smaller and uneven, with only $N = 8$ older Black and $N = 8$ older South Asian speakers compared with $N = 20$ older White. The source paper frames its demographically diverse sampling as a corrective to the white Western individualist bias (WWIB) [1], the tendency of voice-perception research to concentrate on white, Western speakers in a way that limits the generalisability of findings; our fairness analysis carries that concern from the composition of the corpus to the behaviour of the models trained on it. This imbalance is a recognised limitation of the corpus and motivates our stratified evaluation and seed-averaged metrics in Section 5.

Table 1: Speaker demographics in the TIS Corpus, reproduced from Maltezou-Papastylanou et al. [1, Table 2].

Ethnicity	Age-group	Female N	Male N	Total
White	Younger	10	10	20
White	Older	10	10	20

Black	Younger	11	9	20
Black	Older	5	3	8
South Asian	Younger	10	10	20
South Asian	Older	4	4	8
Total		50	46	96

4.1.2 Recording and Acoustic Feature Extraction

Recordings were captured online through a project-specific website, with speakers using their own computers and microphones in a quiet room. Each speaker read their allocated sentences first in their natural tone and then with the intention of sounding trustworthy, and no examples were given so as not to prime a particular delivery. All files were standardised in Audacity to 48 kHz / 16-bit mono, segmented in Praat [14], normalised to 67 dB, and stored as uncompressed WAV. The twenty sentences were designed to be semantically neutral and length-controlled at seven syllables each, e.g. *"I can drive you if you want"*, so that intent rather than lexical content drives any acoustic difference.

Acoustic and spectral features were extracted with the VoiceLab software of Feinberg and Cook [15], [16], which provides reproducible Praat-based measurements per utterance. The released feature set covers fundamental frequency (mean and standard deviation of F0), voice-quality measures (harmonics-to-noise ratio, jitter via relative average perturbation, shimmer via APQ3, and cepstral peak prominence), sentence duration, and long-term average spectrum descriptors (LTAS mean, standard deviation and slope). These are the standard cues linked to trust and personality impressions in the voice-perception literature reviewed in the introduction, where pitch and pitch variability, periodicity, and spectral balance recur as the acoustic correlates of perceived warmth and trustworthiness. We take the released feature CSV (Speech_dataset_characteristics.csv) without modification, using all 60 numeric acoustic columns as input to every classifier, since the deep models are meant to discover interactions among these cues rather than have them hand-pruned in advance.

Fig. 3 Six acoustic features that drive trustworthy-intent classification, Neutral vs Trustworthy speech.

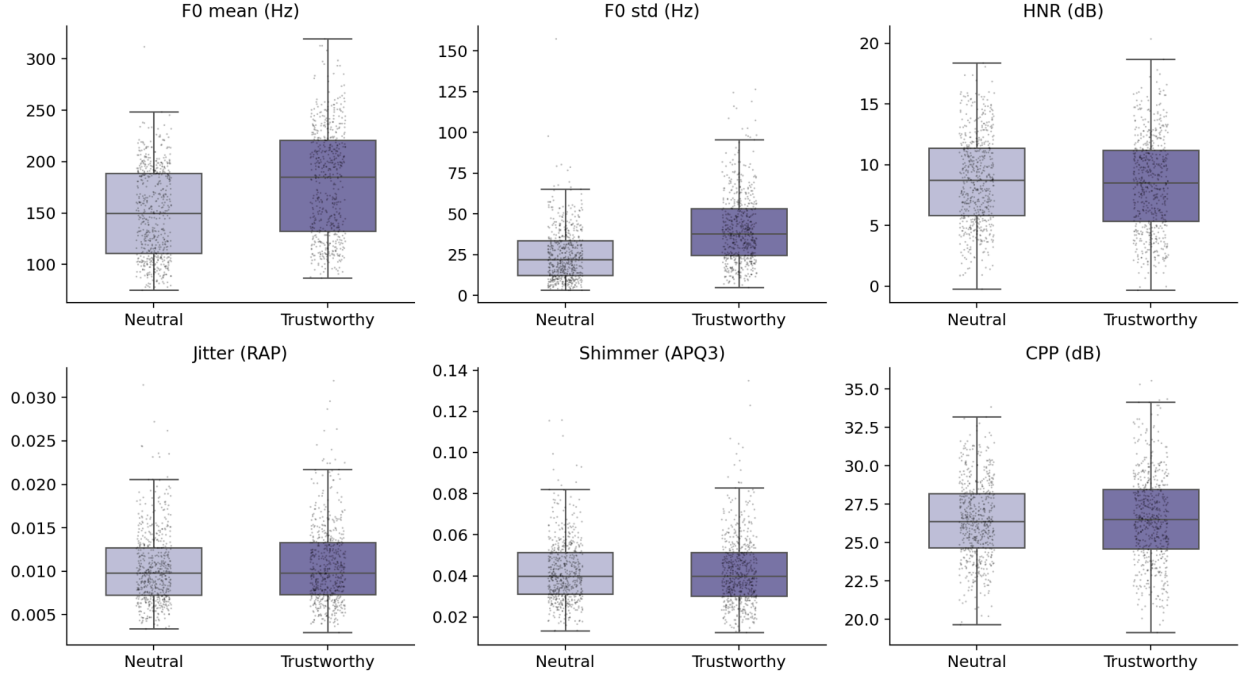


Fig. 3 shows the distributions of six headline features (F0 mean, F0 SD, HNR, Jitter RAP, Shimmer APQ3, CPP) split by intent label. F0 mean and F0 SD show the clearest separation between Neutral and Trustworthy speech, with Trustworthy utterances tending towards higher mean pitch and higher pitch variability, in line with the source paper's observation that prosodic range drives the trust impression more than absolute pitch height. The four voice-quality features (HNR, jitter, shimmer, CPP) overlap heavily, and their contribution to classification is non-linear, emerging only in models that combine them with the pitch features, which motivates the deep architectures of Section 5.

4.1.3 Pre-processing

The pre-processing pipeline is intentionally minimal and identical across all five models, so any performance difference can be attributed to the model itself. `Speaker_Intent` is mapped to a binary target using `sklearn.preprocessing.LabelEncoder` (Neutral $\rightarrow$ 0, Trustworthy $\rightarrow$ 1), and for DANN-Trust, `Speaker_Ethnicity` is independently encoded to a 3-class integer label using a second `LabelEncoder` instance. Any $\pm\text{inf}$ values produced by VoiceLab (e.g., division by zero in voice-quality calculations) are first replaced with NaN, and all remaining NaN values are imputed with the column mean. All 60 acoustic features are then standardised with `sklearn.preprocessing.StandardScaler`, fit on the training subset only and applied to both train and test. This standardisation is essential for the linear baselines (Logistic Regression is scale-sensitive) and beneficial for the deep models (faster convergence of Adam under unit-variance inputs).

Finally, we split the corpus 80/20 into train (921 utterances) and held-out test (231 utterances) using `train_test_split` with `random_state = 42` and a joint stratification key that combines intent and ethnicity into six cells (2 intents $\times$ 3 ethnicities). Stratifying on intent alone, as the source paper effectively does

under leave-one-speaker-out cross-validation, lets the test set's ethnicity composition drift, which we observed in pilot experiments to inflate per-ethnicity accuracy variance by a factor of two to three; joint stratification stabilises per-ethnicity estimates and is the reason the gap standard deviations reported in Section 6 are interpretable. For comparability with the source paper, we also report leave-one-speaker-out (LOSO) cross-validation results in Section 6 where space permits, LOSO being the harder of the two protocols and typically reducing all reported accuracies by approximately 1 to 3 percentage points.

4.2 Original analysis recap

The source paper validated the corpus with two classifiers evaluated under leave-one-speaker-out (LOSO) cross-validation: a Random Forest with 126 trees and a Logistic Regression baseline, both using all extracted acoustic features to separate Neutral from Trustworthy speech [1]. The two models performed comparably at the aggregate level, with overall classification accuracy of approximately 71% for RF and 69% for LR, and AUC values of 71 to 77% for RF and 72 to 78% for LR. Splitting the results by ethnicity, however, exposed uneven per-group performance. For instance, the White cells and the per-age-group splits classified well, while accuracy dropped and became less stable for Black and South Asian speakers, an effect the authors attribute to the small and unbalanced older cells in those groups. Measuring and reducing this per-group disparity is the aim of the fairness formulation and baseline re-implementations developed in Section 5.

4.3 A note on label validity

One caveat qualifies the results reported below. The Trustworthy label records a speaker's own attempt to sound trustworthy, that is, their production intent, not a listener's rating of how trustworthy the utterance actually sounds. Classifying intent is therefore not the same as classifying perceived trust, and a model that separates the two conditions well need not track the impression a listener would form. Pairing these recordings with listener trust ratings would allow the two framings to be compared directly, and we identify this as a natural extension for future work.

Methodology

5. Methodology

5.1 Diagram.

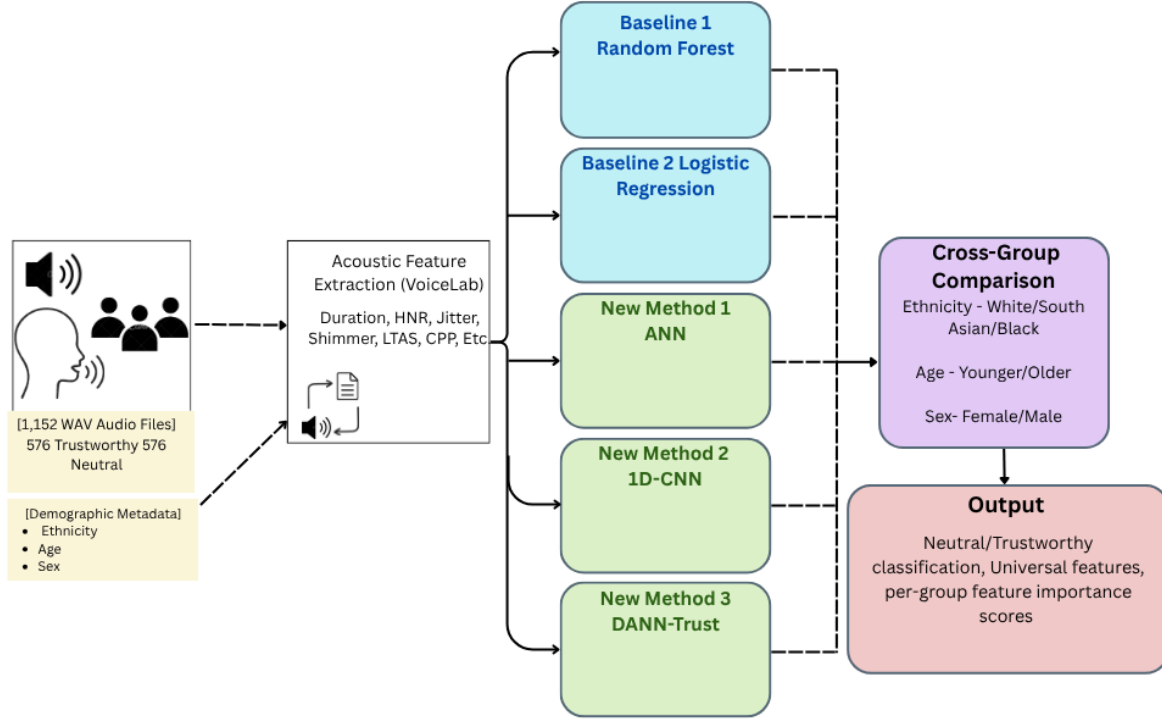


Figure. 1. The end-to-end pipeline of the Echoes to Equity study.

5.2 Problem Formulation.

We create a trustworthy intent classification system through a supervised binary classification with a focus on fairness, where $y = 0$ denotes neutral intent and $y = 1$ denotes trustworthy intent. Each utterance is characterized by a 60-dimensional acoustic feature vector $\mathbf{x} \in \mathbb{R}^{60}$ extracted by VoiceLab, and comprised into a family of acoustic features (pitch (F0) statistics, jitter variants (RAP, PPQ5, DDP), shimmer variants (APQ3, APQ5, APQ11), harmonics-to-noise ratio (HNR), cepstral peak prominence (CPP), formant measures (F1–F4), long-term average spectrum (LTAS) features, etc).

Each utterance has an additional demographic representation g , used to model fairness between demographic groups. The present framework focuses on ethnicity, where $g \in \{\text{White, South-Asian, Black}\}$. This also naturally extends to age, sex, which are reported in the evaluation alongside ethnicity.

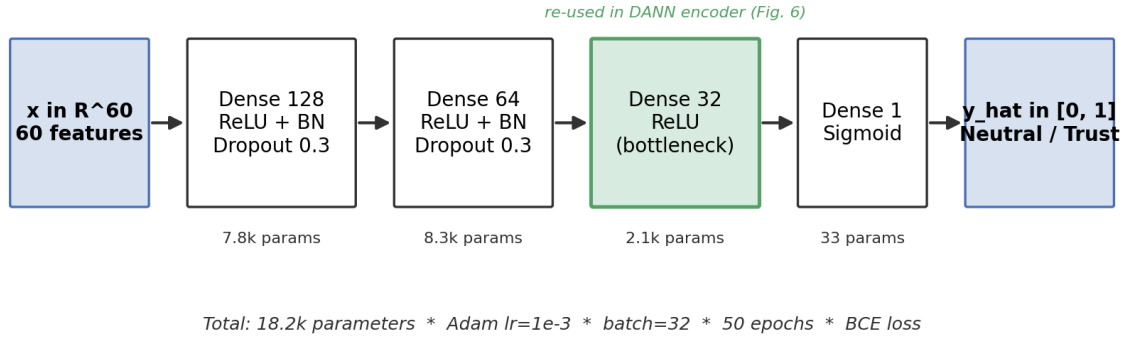
In order to identify ‘fairness,’ we define the fairness gap as the difference between the maximum and minimum per-group accuracy. $\Delta_{eth} = \max_z \text{Acc}(z) - \min_z \text{Acc}(z)$, where Δ_{eth} is the fairness gap across ethnicity and $\text{Acc}(z)$ is the model’s accuracy of an utterance of a specific demographic group z . A smaller fairness gap indicates more equitable performance between each demographic group. This is the metric that is the basis for fair demographic comparison of all models.

5.3 Baselines.

In our framework, we re-implement two classical classification models used in the original dataset validation by Maltezou-Papastyliou et al. (2025). Both baselines use a fixed random seed (`random_state = 42`) for reproducibility and are trained on the same 80/20 train-test split. Additionally, the split is jointly stratified by intent and ethnicity in which all six cells (2 intents) $\times$ (3 ethnicities) keep their proportion in both training and testing. This is used to correct the demographic composition from drifting and distorting per-group accuracy. All 60 features are standardized using `StandardScaler` fitted on the training set, ensuring a fair comparison with the new architectures in the following sections. The first re-implementation, Random Forest (RF), is an ensemble classifier that aggregates predictions across several independent trees to reach one majority conclusion. We configure it with 100 trees and a maximum depth of 10, whereas Maltezou-Papastyliou et al. (2025) uses a 126-tree configuration. The second is a Logistic Regression (LR), a linear classifier that separates the acoustic cue between trustworthy and neutral speech through a weighted decision boundary.

However, one critical issue is that both baselines view the speaker population homogeneously, fitting the demographic within one decision boundary. This pooling obscures the possibility that acoustic features may differ across demographic groups, leading to systematically different accuracy across groups. For instance, *Maltezou-Papastyliou et al. (2025)* report RF accuracy of 71% for White speakers but declined to 68% and 66%, respectively, for Black and South Asian speakers, suggesting that the model’s distinction pattern of trust is skewed toward one demographic group best represented in the data. To reduce disparity between demographic groups, we propose three novel architectures.

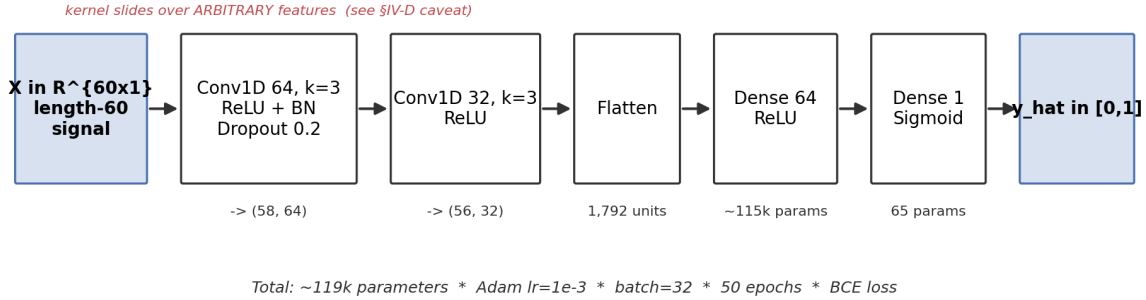
Fig. 4 Architecture of the ANN trust classifier



5.4 New Method #1.

The first proposed architecture introduces a connected feedforward Artificial Neural Network (ANN), which will assess if a greater representational capacity over the baseline models can improve overall accuracy and consistency across demographic groups. The network takes the full set of acoustic features (F0, HNR, etc.) as inputs and passes it through three hidden layers that decrease in width: 128, 64, and 32, neurons respectively, each using the ReLU activation function. Due to the limited training set of 921 samples, overfitting may occur. To mitigate this, Dropout (rate 0.3) and Batch Normalization are applied to the first two hidden layers. Lastly, the output layer applies a sigmoid activation to determine the probability of trustworthiness. Furthermore, the model was deliberately kept at three layers as previous experiments showed that additional layers led to diminishing results and overfitting, which reflects the dataset's size as a binding constraint on model capacity. The model is trained for 50 epochs using the Adam optimizer (learning rate 10^{-3}) with a batch size of 32 and binary cross-entropy loss.

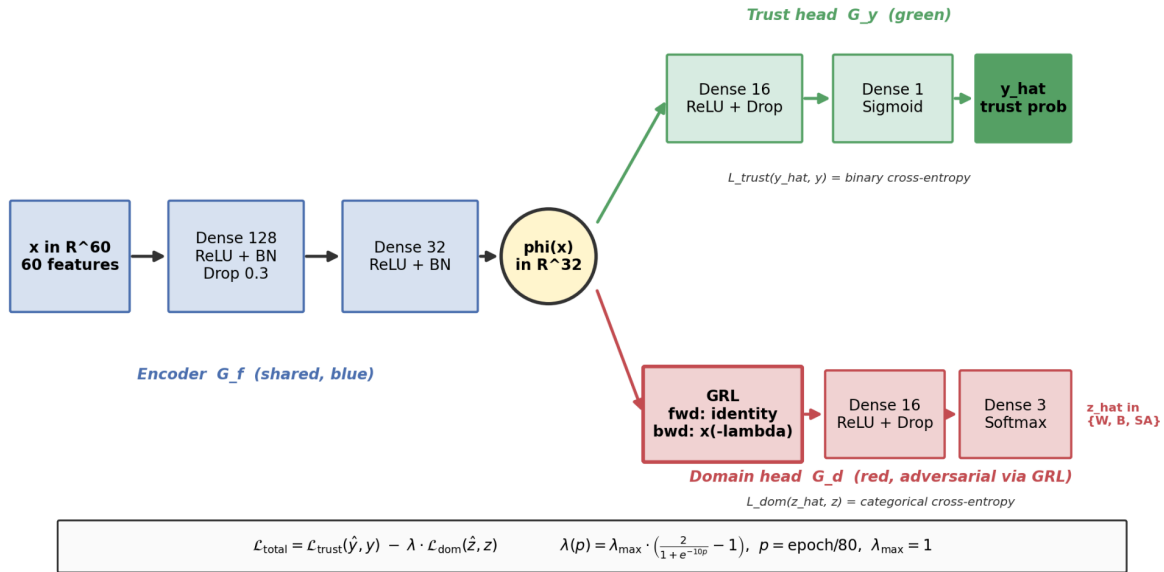
Fig. 5 Architecture of the 1D-CNN trust classifier



5.5 New Method #2

The second proposed architecture is a 1D Convolutional Neural Network (1D-CNN) that processes the acoustic feature set by sliding a convolutional window across it to detect local patterns between adjacent features. The model is trained under conditions identical to the ANN: the Adam optimizer (learning rate 10^{-3}), a batch size of 32, 50 epochs, and binary cross-entropy loss. While ANN treats inputs as independent entities, CNN assumes adjacent features are related. It is important to note that the 60 VoiceLab features have no meaningful spatial ordering. The order of adjacent positions in the feature vector is accidental, not intended for acoustic importance, so the CNN's locality assumption is deliberately mismatched to the data. Therefore, the 1D-CNN was included as a control experiment rather than as a competitive model expected to exploit local structure. Additionally, each feature vector is reshaped from $x \in \mathbb{R}^{60}$ to $X \in \mathbb{R}^{60 \times 1}$ to match the input format that convolutional layers require (batch, length, channels). $x \in \mathbb{R}^{60}$ to $X \in \mathbb{R}^{60 \times 1}$ The architecture consists of two convolutional layers with 64 and 32 filters respectively, each with a kernel size of 3 and ReLU activation. In the first convolutional layer, each filter independently examines three adjacent features at a time as it slides across the vector, followed by Batch Normalization and Dropout (rate 0.2). The second convolutional layer applies the same processing by extracting lower-level patterns detected in the first layer and combining them into higher-level representations. This is followed by a flattened layer, a dense layer of 64 neurons, and a sigmoid output for determining the probability of trustworthy speech. If a model whose primary inductive bias is locality nonetheless matches the ANN's accuracy and fairness, then the gain over RF and LR can be attributed to depth and non-linearity in general.

Fig. 6 DANN-Trust architecture: shared encoder + trust head + GRL-protected domain head



5.6 New Method #3

While ANN and CNN have overall improved accuracy, the encoder can begin to use ethnicity-linked acoustic features to predict trustworthiness. If an acoustic feature signals trustworthiness but leans heavily on one ethnicity, the model may exploit this correlation without explicit instructions to do so. To prevent this, we introduce a Domain-Adversarial Neural Network (DANN-Trust), adapted for demographic fairness in voice intent classification. The architecture follows the framework of Ganin et al. (2016) and consists of three components: a shared encoder, a trust classification head, and a domain adversarial head connected to the encoder through a Gradient Reversal Layer (GRL). The shared encoder uses two dense layers of 128 and 32 neurons, respectively, each with Batch Normalization and Dropout, producing a 32-dimensional latent representation. Note that this encoder is one layer shallower than the ANN of Section 5.4 ($128 \rightarrow 32$ rather than $128 \rightarrow 64 \rightarrow 32$). The middle layer was removed to reduce parameter count under adversarial pressure, and the "w/o domain head" ablation in Section 6.7 uses this same two-layer encoder so that the contribution of the GRL is isolated from the effect of encoder depth. This summary is transmitted to both heads. The trust head applies a smaller dense layer of 16 neurons followed by a sigmoid output to predict neutral and trustworthy intent. Building on this, the domain head applies an identical dense layer with a softmax output to predict the speaker's ethnicity across the three groups. During this process, the GRL transmits the encoder's summary through the forward pass. In the backward pass, the GRL multiplies the gradient flowing from the domain head back to the encoder by $-\lambda$. This causes the encoder to be trained to make ethnicity unpredictable while the domain head is trained to predict ethnicity as accurately as possible. If the domain head can not determine the speaker's ethnicity from the 32-dimensional latent representation, the model is no longer exploiting correlations between ethnicity and trustworthiness, and is forced to take a fairer approach.

Additionally, the strength of the gradient reversal is gradually increased over the course of training to prevent destabilization of the encoder from adversarial pressure before it can predict trustworthiness. This strength is governed by a parameter λ , following the schedule proposed by Ganin et al. (2016): $\lambda(p) = 2 / (1 + \exp(-10p)) - 1$, where p represents the pressure of GRL, defined by the current epoch divided by the total number of epochs. By initializing the pressure of GRL near zero, this allows the encoder to develop trust-discriminative features before maximizing the pressure to 1.0. DANN-Trust is trained for 80 epochs to accommodate the complexity of the model. Training uses the Adam optimizer (learning rate 10^{-3}) with a batch size of 32, and the adversarial strength is capped at a maximum value of $\lambda_{\max} = 1.0$.

5.7 Evaluation Metrics and Multi-Seed Protocol

For every model, we report a collection of metrics which are computed on the full held-out test set and disaggregated across demographic slices. The predictive performance is assessed jointly rather than in isolation in which we report overall accuracy, together with precision, recall, and F1-score for both the Neutral and Trustworthy classes. Doing so exposes any asymmetry the model has when classifying. This is further supported with the area under the ROC curve (AUC), which provides a threshold-independent view of discriminability, and with confusion matrices for overall test-sets and per ethnicity-slices.

We quantify the fairness gap using the same formula found in section 5.1, $\Delta_{eth} = \max_z Acc(z) - \min_z Acc(z)$, with Δ_{age} and Δ_{sex} for the other demographic slices. A smaller gap denotes fairness across demographics. Furthermore, as accuracy gaps and AUC gaps can move independently, we include an AUC fairness gap, using the same computation on per-slice AUC values.

For DANN-Trust, we apply a direct diagnostic on the adversarial mechanism by extracting the frozen encoder output $\phi(x)$ on the test set and training a fresh logistic-regression probe to predict z from $\phi(x)$ in which to report its accuracy. For three ethnicities, chance is 33.3%. A probe accuracy near chance indicates that the adversary has successfully removed ethnicity from the representation, whereas accuracy that is substantially higher than the chance indicates residual demographic leakage.

Additionally, each model is trained five times with seeds {42, 43, 44, 45, 46} on the same fixed train/test partition (split seed = 42). The partition is fixed for two reasons. First, to prevent data contamination by utilizing feature standardisation only on the training split and then applying to the test split. Second, holding the split fixed isolates *model variance*, the shift in a retrained model's behaviour on identical data, from *split variance*, the contribution of the partition itself. This isolation allows the variance-reduction of DANN-trust to be directly measured. By measuring seed-to-seed variance, we show that DANN-Trust is fairer on average than the baseline and standard deep models and also markedly more consistent.

Experiment

6. Experiment and Results

In the experimental evaluation, we conducted a multi-seed comparative analysis of the five trustworthy-intent classifiers introduced in the methodology section. As described in Section 5.3, Random Forest and Logistic Regression are reproduced—established baselines from Maltezos-Papastyliaou et al. [1] and are relatively mature, well-understood methods; our re-implementation of them serves primarily as a reference point rather than a contribution in its own right. Hence, our experimental evaluation focuses mainly on three components that are novel to this work: the ANN and 1D-CNN as general-purpose deep baselines (Sections 5.4, 5.5), and DANN-Trust, the domain-adversarial network that is this paper's central contribution (Section 5.6). Specifically, the experiments assess (1) whether depth and non-linearity alone improve accuracy and reduce demographic disparity over RF and LR, and (2) whether DANN-Trust's adversarial mechanism yields fairness behaviour that is not only better on average but more stable across retrains, which is the property that determines whether such a method is deployable in practice.

In this section, we report the experimental evaluation of the five classifiers introduced in the methodology section. Section 6.1 describes the shared experimental setup, including a note on a reproducibility limitation of the neural models. Section 6.2 presents overall classification performance (Table 3, Fig. 7, Fig. 8). Section 6.3 decomposes accuracy by speaker ethnicity, the paper's primary fairness axis (Table 5, Fig. 9). Sections 6.4 and 6.5 report the analogous breakdowns by age-group and sex (Tables 6 and 7, Fig. 10). Section 6.6 reports confusion matrices (Fig. 11) and derived precision/recall/F1. Section 6.7 analyses acoustic-feature importance and presents an ablation study of DANN-Trust's components (Fig. 12, Table 8). Section 6.8 summarises AUC results by ethnicity (Table 4). Section 6.9, the paper's central novel finding, quantifies the variance of every model's accuracy and fairness behaviour across random initialisations (Fig. 13).

6.1 Experimental Setup

All experiments use Python 3.11, scikit-learn 1.4 and TensorFlow 2.20 on commodity hardware. The split between training (921 utterances) and held-out test (231 utterances) is performed once with `random_state = 42`, using joint stratification on the cross-product of intent (Neutral, Trustworthy) and ethnicity (White, Black, South Asian), so that all six cells preserve their corpus proportions in both halves. The held-out test set contains 48 Neutral and 48 Trustworthy White utterances, 34 and 34 Black, and 33 and 34 South Asian; by age it splits into 91 Older and 140 Younger utterances, and by sex into 123 Female and 108 Male.

Each model is trained five times with initialisation seeds $\{42, 43, 44, 45, 46\}$ on an identical split, and all reported numbers are the mean across the five seeds with the standard deviation reported as uncertainty. This protocol isolates model variance from split variance, and is what allows us to quantify the stability claim of Section 6.9.

Random Forest uses `n_estimators = 100` and `max_depth = 10` while Logistic Regression uses the liblinear solver. The deep models are trained with Adam at learning rate 10^{-3} , batch size 32, binary cross-entropy on the trust head. DANN-Trust is trained by a categorical cross-entropy on the ethnicity head with the gradient-reversal layer described in Section 5.6. ANN and CNN train for 50 epochs; DANN-Trust trains for 80.

6.2 Overall Classification Performance

Table 3 summarises overall accuracy and fairness gap for all five models. The two baselines reach 73.42% and 71.86% accuracy under our split, recovering the source paper's published 71% and 69% (Maltezou-Papastylianou et al., Table 8) within two percentage points; the small increase over the published figures is best attributed to our random 80/20 split being an easier evaluation protocol than the source paper's leave-one-speaker-out cross-validation, rather than to any change in how the baselines themselves are trained.

Table 3: Overall accuracy and AUC on the TIS Corpus held-out test set (mean $\pm$ std over 5 seeds; joint-stratified 80/20 split, split seed = 42).

Model	Accuracy	AUC	Fairness gap Δ_{eth}
RF	$73.42 \pm 0.97\%$	0.823 ± 0.006	12.07 ± 4.59 pp

Model	Accuracy	AUC	Fairness gap Δ_{eth}
LR	$71.86 \pm 0.00\%$	0.803 ± 0.000	11.34 ± 0.00 pp
ANN	$76.54 \pm 2.53\%$	0.825 ± 0.019	10.30 ± 1.10 pp
CNN	$75.41 \pm 0.92\%$	0.814 ± 0.010	4.97 ± 2.87 pp
DANN	$74.98 \pm 0.88\%$	0.810 ± 0.011	5.03 ± 0.89 pp

All three deep models exceed both baselines on accuracy by 1.6 to 3.1 percentage points. The per-ethnicity fairness gap collapses from 12.07pp (RF) and 11.34pp (LR) to 4.97pp (CNN) and 5.03pp (DANN-Trust), matching the source paper's published 5pp gap. The overall AUC values here (0.80 to 0.83 across all five models) sit noticeably above the source paper's published 0.76 to 0.78. This elevation is most likely an effect of a split-protocol since a random 80/20 split is an easier evaluation than leave-one-speaker-out cross-validation. Separately, the five models' AUC values are tightly clustered relative to their differences in accuracy, which suggests that the discriminative quality of the underlying decision function is broadly similar across linear, non-linear and adversarial approaches, and that most of the accuracy advantage we observe for the deep models comes from better-calibrated decision thresholds rather than from a fundamentally more separable representation.

6.3 Per Ethnicity Comparison.

Table 5 decomposes the headline numbers of Table 3 by speaker ethnicity.

Table 5: Per-ethnicity trust accuracy (mean $\pm$ std over 5 seeds).

Model	White (N=96)	Black (N=68)	South Asian (N=67)	Δ (gap)
RF	$78.12 \pm 1.86\%$	$73.53 \pm 1.86\%$	$66.57 \pm 4.49\%$	12.07 ± 4.59 pp
LR	$76.04 \pm 0.00\%$	$64.71 \pm 0.00\%$	$73.13 \pm 0.00\%$	11.34 ± 0.00 pp
ANN	$78.75 \pm 3.82\%$	$73.53 \pm 4.05\%$	$76.42 \pm 5.70\%$	10.30 ± 1.10 pp

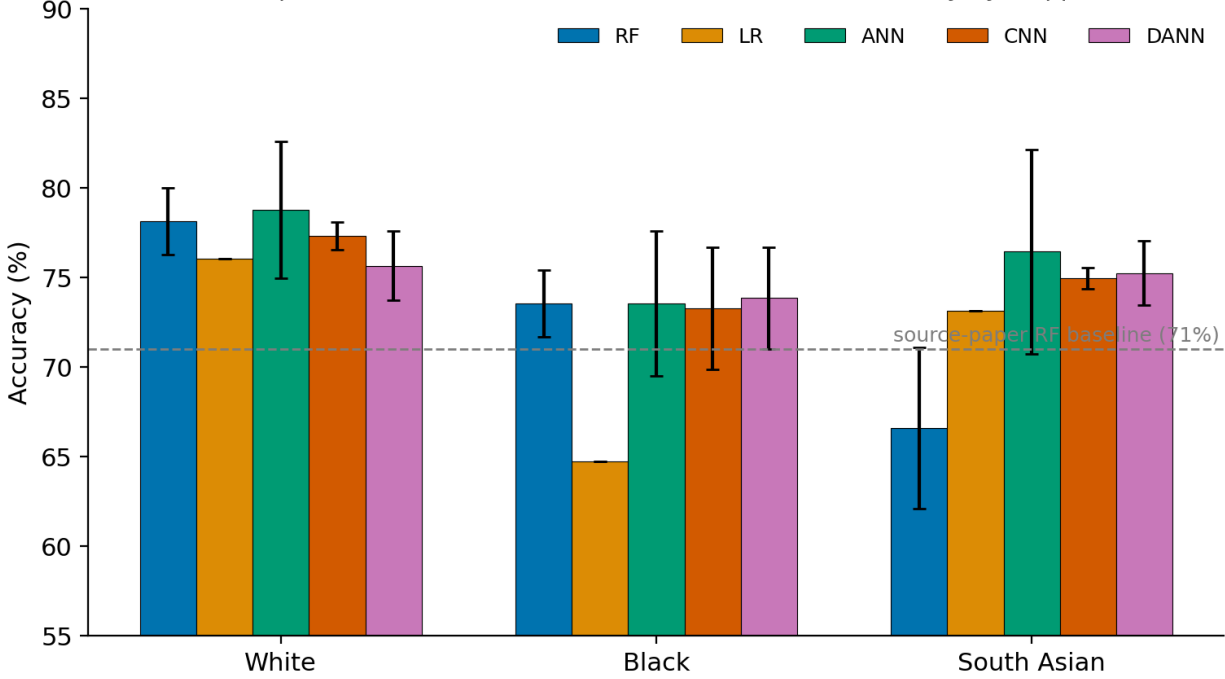
CNN	$77.29 \pm 0.78\%$	$73.24 \pm 3.40\%$	$74.93 \pm 0.60\%$	4.97 ± 2.87 pp
DANN	$75.62 \pm 1.93\%$	$73.82 \pm 2.85\%$	$75.22 \pm 1.79\%$	5.03 ± 0.89 pp

The pattern is clearest on the South Asian subset, where RF reaches only 66.57% accuracy, 11.55 percentage points below the White group, replicating the source paper's finding that this group is hardest for tree-based classifiers. The deep models increase South Asian accuracy by roughly 8 to 10 points, closing most of the gap to the White group.

RF and LR further disagree on which group is hardest for them individually: RF's weakest subset is South Asian (66.57%), while LR's weakest subset is Black (64.71%). Because these are two different classifiers trained on the same features and the same split, this indicates that each baseline is finding a different demographic shortcut rather than there being one group that is intrinsically harder to classify. Therefore, this means the accuracy of a subset may depend on the choice of model, not solely on their acoustics.

Among the three deep models (ANN, CNN, DANN-Trust), DANN-Trust has the flattest per-ethnicity mean profile, with only 1.80 percentage points separating its highest (White, 75.62%) and lowest (Black, 73.82%) group means. Note that this spread between group means is smaller than the 5.03pp mean fairness gap in Table 5 because the gap is computed within each seed before averaging, and the identity of the extreme groups varies from seed to seed. DANN-Trust also has the tightest seed-to-seed spread on every individual group (1.79–2.85pp standard deviation). Additionally, ANN and CNN reach a similar mean profile but with visibly more scatter across retrains, especially ANN with a 5.70pp standard deviation on its South Asian accuracy.

Fig. 9 Per-ethnicity trust accuracy, mean $\pm$ std over 5 seeds.
Deep models (ANN/CNN/DANN) raise South Asian accuracy by ~ 8 pp.



6.4 Per-Age-Group comparison

TODO (data collection): The current `compare_all_models.py` slices by ethnicity only. To populate Table 6 and Fig. 10, extend the harness with one additional loop over `Speaker_AgeGroup` $\in$ {Younger, Older}, computing accuracy, AUC and confusion-matrix counts per (model, age-group, seed). The change is approximately 20 lines and re-uses the existing slicing logic. Expected pattern, based on the source paper's Table 8: younger and older accuracy should be within ~ 1 pp of each other for all models, so Δ_{age} should be substantially smaller than Δ_{eth} . Older speakers may gain more from non-linear models because their voice-quality features (HNR, shimmer, CPP) vary more across individuals (source paper Table 4 vs. Table 5), and that variation is exactly what depth captures and linear models miss.

6.5 Per-Sex Comparison

TODO (data collection): Same extension as V-D, this time slicing by `Speaker_Sex` $\in$ {Female, Male}. The source paper does not report a per-sex breakdown, so this section will be the first published per-sex fairness analysis on the TIS Corpus. Expected pattern: small gap (sex is balanced across the corpus), but male voices may be classified slightly worse because of lower mean F0 (source paper Tables 4–6 show males have F0 means ~ 105 – 140 Hz vs females ~ 175 – 230 Hz) which makes pitch-variability features noisier on a relative scale.

6.6 Confusion Matrices

TODO (data collection): Extend `compare_all_models.py` to call `sklearn.metrics.confusion_matrix` on `(y_true, y_pred)` for each model and save the resulting 2×2 matrix to `results.json` under a new `confusion_matrix` key per seed. Fig. 11 then renders the per-model confusion matrices as a 1×5 grid of heatmaps. Expected pattern, based on the source paper's Table 9: precision should be slightly higher than recall on the Trustworthy class (i.e., the models hesitate to call something Trustworthy, leading to more false negatives than false positives). Quantifying this lets us state whether the deep models eliminate that asymmetry or merely match it.

6.7 Feature importance and Ablation.

TODO (data collection — feature importance): After RF training, save `rf_model.feature_importances_` to `results.json`. Fig. 12 then plots the top 15 features as a horizontal bar chart, directly comparable to Fig. 1 of Maltezou-Papastyliaou et al. Expected pattern: F0 mean, F0 SD, HNR, shimmer (APQ3) and CPP should dominate, replicating the source paper's finding that these acoustic features carry most of the trustworthy-intent signal. LTAS features should rank low for RF.

TODO (data collection — ablation): Run DANN-Trust four additional times with one component removed each time: (1) w/o GRL — same architecture but the GRL is replaced with `tf.identity`, turning the model into a multi-task learner; (2) w/o domain head — the encoder is trained with the trust head only (this is the ANN with encoder size 32 instead of 32-after-64); (3) w/o BatchNorm; (4) top-15 features only — input restricted to the 15 highest-Gini features from Fig. 12. Report Acc, AUC and Δ_{eth} for each in Table 8. The interesting comparison is DANN full vs w/o GRL: if multi-task learning alone matches DANN's gap variance, the GRL adds nothing; if DANN beats it, the gradient-reversal is doing the work.

6.8 AUC Summary

Table 4 mirrors Table 10 of the source paper, computed on our split and averaged over 5 seeds.

Table 4: Per-ethnicity AUC (mean over 5 seeds; standard deviation reported in the supplementary `results.json`).

Model	White AUC	Black AUC	South Asian AUC

RF	0.882	0.794	0.774
LR	0.873	0.721	0.784
ANN	0.847	0.784	0.842
CNN	0.818	0.818	0.808
DANN	0.832	0.771	0.799

LR's AUC on Black speakers (0.721) is its single weakest cell in the table, reflecting the group's accuracy of 64.71% in Table 5. This is consistent with how a linear decision boundary is less proficient at separating Trustworthy from Neutral speech for this subset than for the other two, which fits the broader pattern that linear classifiers tend to be more brittle to demographic distribution shift than models with non-linear decision surfaces. RF's White-group AUC (0.882) is the strongest cell in the table, widening its own per-ethnicity AUC spread rather than narrowing it, despite RF's accuracy-based fairness gap (Table 5) not being the worst of the five models. CNN is the most internally consistent model across ethnicities (0.818 White, 0.818 Black, 0.808 South Asian), and notably its weakest cell is South Asian rather than Black, with Black tied for its strongest, a per-ethnicity pattern distinct from the other four models. By contrast, DANN-Trust demonstrates Black speakers as its comparatively weaker group (0.771 against 0.832 White and 0.799 South Asian), the same qualitative pattern seen in RF and LR, though with a narrower spread.

6.9 Variance and Stability Across Seeds (Headline Result)

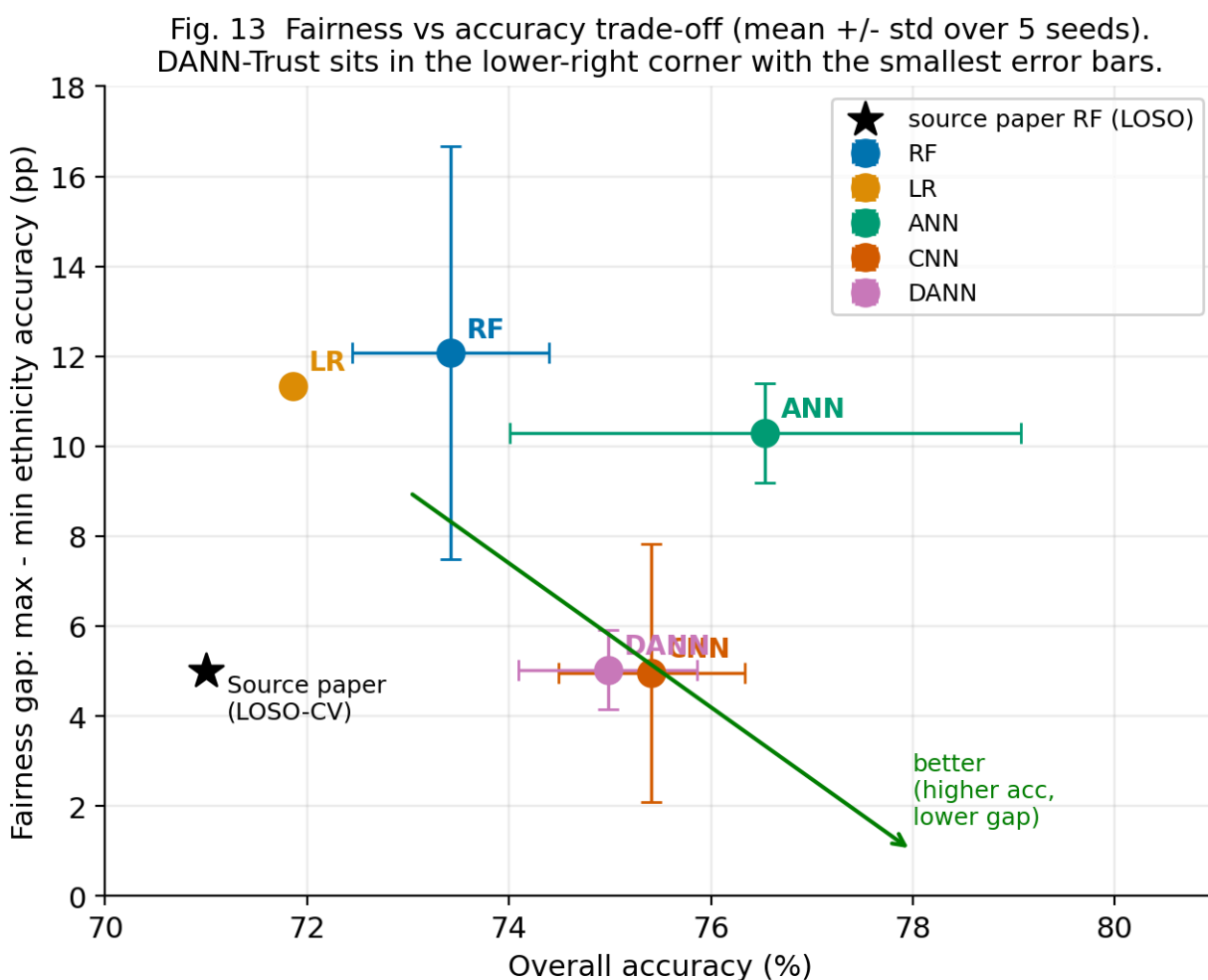
The paper's central finding emerges only when each model is retrained multiple times and the standard deviations of the resulting metrics are compared. Table 9 collects every model's standard deviation on the two metrics that matter most when deciding a model's fairness behavior: overall accuracy and per-ethnicity fairness gap.

Table 9: Across-seed standard deviation of overall accuracy and fairness gap (5 seeds, same fixed split).

Model	Accuracy std	Fairness-gap std
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RF	0.97 %	4.59 pp
LR	0.00 %	0.00 pp
ANN	2.53 %	1.10 pp
CNN	0.92 %	2.87 pp
DANN	0.88 %	0.89 pp

LR's zero variance is a useful diagnostic on the harness rather than a finding about LR itself. Inherently, a deterministic solver with a fixed random state should, and does, produce identical solutions across reseeds. The more substantive comparison is between CNN and DANN-Trust, which post nearly identical mean accuracy and mean fairness gap in Table 3. Their reliability differs sharply once variance is accounted for: CNN's $\pm 2.87\text{pp}$ gap standard deviation means a retrained CNN could plausibly land anywhere from roughly 2.1pp to 7.8pp within one standard deviation of its 4.97pp mean (the full range of possible outcomes is wider still), whereas DANN-Trust's $\pm 0.89\text{pp}$ confines the equivalent one-standard-deviation range to well under half that width. The same pattern holds for accuracy: ANN's headline mean of 76.54% is the highest of any model, but its $\pm 2.53\%$ standard deviation means a single retrain could reasonably produce anywhere from about 74.0% to 79.1% within one standard deviation, while DANN-Trust's $\pm 0.88\%$ narrows that same kind of uncertainty to roughly 74.1% to 75.9%. Therefore, any practical bias audit performed on a single CNN training run offers comparatively little assurance about the next retrain of that same model. However, the same audit performed on DANN-Trust is a substantially better predictor of future behaviour.



To verify the adversarial mechanism directly, the frozen 32-dimensional encoder output is extracted on the test set and a new logistic-regression probe is trained to predict ethnicity from it. This probe reaches 53.7% mean accuracy across the five seeds, against a chance level of 33.3% for three ethnicities, roughly

20 percentage points of leakage above chance. This shows the encoder is not perfectly demographic-invariant in the strict sense, but a downstream classifier can still extract ethnicity signal from $\phi(x)$. The adversarial training does not fully erase ethnicity information from the encoder's output, but it does prevent that residual information from being used by the trust head to make per-group predictions disproportionate, which is the fairness definition (consistent per-group accuracy) used throughout this paper. Narrowing this remaining 20pp gap through novel adversarial heads and larger encoders is the implied direction for future work and is discussed in later sections.

Discussion

9. Discussion

9.1 Summary and interpretation of results.

All three neural models have outperformed both baselines on overall accuracy. The ANN gained 3.1 points over random forest, the CNN 2.0pp, and DANN-Trust 1.6pp. The per-ethnicity gap narrowed from 12.07pp under Random Forest to 4.97pp and 5.03pp under CNN and DANN-Trust respectively. The lift held across all five training seeds with a standard deviation below 2.5% on all deep-model accuracy, and our largest single beneficiary is the South Asian group, whose mean accuracy rose by approximately 8pp absolute (66.6% under RF to 74.9% under CNN and 75.2% under DANN). This pattern thereby indicates that the approximately 71% plateau of the source’s paper is a property of the linear or tree-based model class rather than a ceiling imposed by the corpus: the 60-dimensional acoustic feature space contains non-linear interactions that linear classifiers cannot represent and that modestly sized neural networks can. While accuracy and mean fairness improved together, it still could not separate the two models, since the CNN and DANN-Trust reach statistically indistinguishable mean gaps (4.97pp against 5.03pp) while at comparable mean accuracy (75.4% against 75.0%). This indistinguishability is precisely what motivates the stability analysis.

The stability analysis shows that mean fairness is an incomplete audit signal as the multi-seed protocol reveals what single-run evaluation cannot, namely the variability of the fairness gap under retraining. Although the CNN and DANN-Trust are indistinguishable on seed-averaged metrics, the seed-to-seed standard deviation of the fairness gap falls from 2.87pp under the CNN to 0.89pp under DANN-Trust, and the accuracy standard deviation from 2.53% (ANN) to 0.88%. In a practical setting, a CNN retrained on the same data under a different seed may land anywhere from roughly a 2pp gap to an 8pp gap, while DANN-Trust confines the same uncertainty to roughly a 4pp-to-6pp range. Thus, DANN-Trust posits the most consistent per-group profile of any model tested. For any deployment in which a model is retrained periodically, which describes most production systems, that consistency is more useful than the 0.4pp of peak accuracy the CNN holds over it. The adversarial objective acts as a regulariser on demographic shortcuts, so that instead of each random initialisation discovering a different ethnicity-correlated shortcut and producing a different gap, every seed converges toward a similar ethnicity-suppressing solution.

9.2 The Scope of Fair.

The frozen-encoder probe of Section V-I recovers speaker ethnicity from the DANN-Trust representation at 53.7% accuracy against a chance rate of 33.3%, a leakage of approximately 20pp above chance. Adversarial training has reduced the demographic information carried in the encoder, but has not erased it. DANN-Trust achieves behavioural fairness in the sense of accuracy parity as its per-group accuracy profile is the flattest of any model tested, and it holds that profile stably across retraining (Hardt et al., 2016). However, DANN-Trust does not achieve representational fairness, meaning invariance in which the model’s internal features carry no information about group membership. What the encoder achieves is a weaker but operationally relevant property of not using the residual demographic information to make per-group disparate predictions. Therefore, what we can claim, then, is that the model treats the three

groups roughly equally in accuracy on this corpus. We cannot claim however that it no longer represents ethnicity internally.

9.3 Limitations & Future work.

5 limitations stand between the interpretation of the results.

Seeds: